

NOS Practice Problem Statements

Spec-driven development practice — Intermediate

How this works: pick a problem statement from, write a short spec before writing any code, build a working prototype against that spec, then walk through a quick live demo.

Intermediate

Two related entities with a computed relationship (a split, a schedule conflict, a workflow state). Focus: spec captures relationships and edge cases, not just features.

1. Expense Splitter

Required features:

- Add group expenses
- Split evenly or by custom percentage
- Show who-owes-whom summary

Edge case to handle:

What happens when percentages entered for a custom split don't add up to 100%?

2. Meeting Room Booker

Required features:

- List available rooms
- Book a time slot
- Prevent double-booking
- Show today's schedule per room

Edge case to handle:

What happens when two people try to book the same slot at the same moment?

3. Budget Planner

Required features:

- Set monthly budget by category
- Log expenses against categories
- Show remaining balance per category

Edge case to handle:

What happens when an expense pushes a category balance below zero?

4. Job Application Tracker

Required features:

- Add applications with status (applied/interview/offer/rejected)
- Filter and sort by status
- Show days-since-applied

Edge case to handle:

How should the tracker handle an application that moves backward in status (e.g., interview to rejected)?

5. Subscription Manager

Required features:

- Track subscriptions with renewal dates
- Show monthly total spend
- Flag upcoming renewals within a 7-day window

Edge case to handle:

What happens to a subscription's renewal flag if the user opens the app after the renewal date has already passed?

6. Team Task Board

Required features:

- Tasks with assignee and status (todo/in-progress/done)
- Show per-person workload count
- Flag overdue tasks

Edge case to handle:

How is workload counted for a task with no assignee — excluded, or shown in an 'unassigned' bucket?

7. Event RSVP Tracker

Required features:

- Set event capacity
- Collect RSVPs
- Waitlist once capacity is full
- Auto-promote from waitlist on cancellation

Edge case to handle:

What happens if the same person tries to RSVP twice?

8. Simple Approval Workflow

Required features:

- Submit a request
- Manager approves or rejects
- Maintain status history log
- Lock edits after approval

Edge case to handle:

What happens if someone tries to edit a request after it's already been approved?